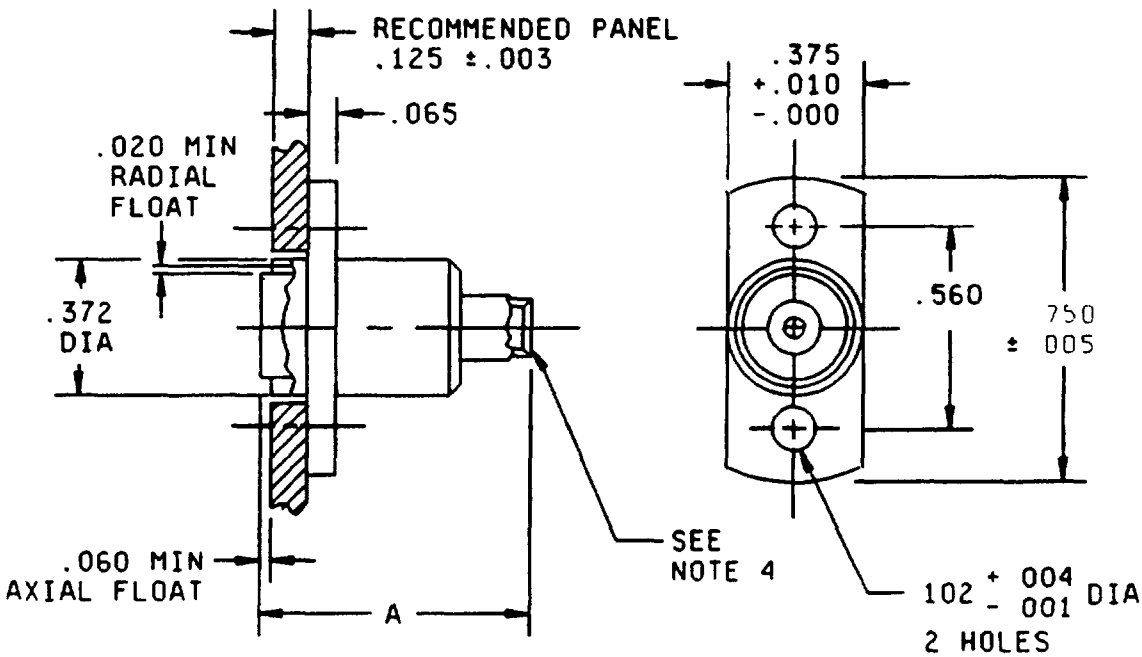
MILITARY SPECIFICATION SHEET

CONNECTOR RECEPTACLE ELECTRICAL RADIO FREQUENCY SERIES BMA FLANGE MOUNT FOR FLEXIBLE CABLE SOCKET CONTACT HIGH RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the latest issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-31031



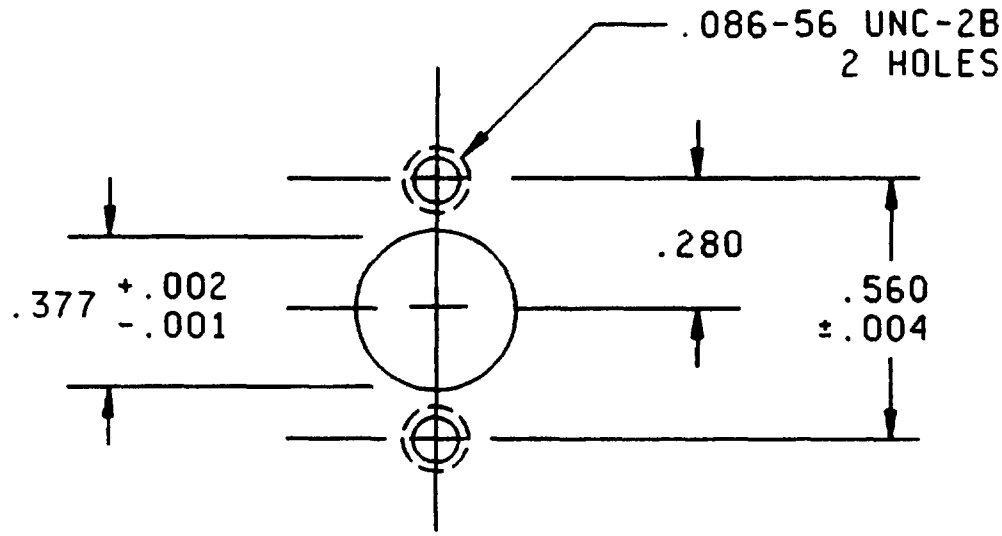
Inches	mm	Inches	mm	Inches	mm
000	0 00	010	0 25	125	3 18
001	0 03	020	0 51	372	9 45
003	0 08	060	1 52	375	9 53
004	0 10	065	1 65	560	14 22
005	0 13	102	2 59	750	19 05

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for information only
3. Unless otherwise specified, all tolerances are ± 005 inch
4. See table I for cable accommodation
5. For dimension A see table I
6. Dimension A defines the overall length of the connector when assembled to the cable

FIGURE 1 General configuration

DISTRIBUTION STATEMENT A Approved for public release distribution is unlimited

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Inches	mm
001	0 03
002	0 05
004	0 10
086	2 18
280	7 11
377	9 58
560	14 22

RECOMMENDED MOUNTING HOLE

Spring compression force and travel The face of the push-on interface shall have a total travel capability of 060 inch toward the connector retainer clip Force versus displacement shall be as follows

Displacement inches	Force pounds
010	3 0 ± 5
050	3 7 ± 5
060	3 8 ± 5

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for information only

FIGURE 2 Recommended mounting hole

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TABLE I PIN s, cross-reference and dimensions

PIN 1/ M31031/01	Applicable cable and group 2/ M17/	Dimensions	Inches (millimeters) maximum
Category A - Field serviceable (no special tools required)			
A**01	113-RG316 group II	A	1 4 (35 56 mm)
A**02	128-RG400 group VI	A	1 4 (35 56 mm)

1/ These connectors have captivated center contacts

2/ The latest version of the cable is applicable

TABLE II Cable retention force

Cable dielectric outer diameter	Pounds (Minimum)	
	Single braid	Double braid
Inches (maximum)		
036	10	N/A
067	20	N/A
110	30	N/A
122	40	45

TABLE III Group qualification 1/2/3/4/

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
I	M31031/01A**02	M31031/01A**01 M31031/01A**02

* Options as defined in the PIN construction

1/ Individual connectors other than those listed are self qualifying

2/ Qualification of connectors qualifies connectors of the same body materials only

3/ Qualification of connectors qualifies connectors of the same environmental construction only

4/ Changes in plating (body or center contact) shall require the corrosion and contact resistance tests be performed

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ENGINEERING DATA

Nominal impedance 50 ohms ±1 ohm

Frequency range 0 to 18 GHz (frequency limit is dependent on cable used)

Voltage rating

Cable group II 250 V rms at sea level 65 V rms at 70 000 feet

Cable group VI 335 V rms at sea level 85 V rms at 70 000 feet

Temperature rating -65°C to +125°C

REQUIREMENTS The following tests assure connector integrity within typical operating conditions and applications Alternate methods shall be made known to the qualifying authority prior to performance of the test The test methods described in the general specification and herein are proven methods and shall be the refree method in cases of dispute

Dimensions and configuration See figures 1 and 2 and MIL-STD-348

Force to engage and disengage

Longitudinal force 3 pounds maximum

Torque Not applicable

Radial misalignment ± 020 inch minimum

Axial misalignment 060 inch minimum

Coupling proof torque Not applicable

Mating characteristics

Reference MIL-STD 348 for dimensions

Hermetic seal Not applicable

Permeability Applicable

Leakage (pressurized connectors) Not applicable

Insulation resistance Method 3003 of MIL-STD 1344 5 000 megohms minimum

Center contact retention 6 pounds minimum axial force

Radial torque Not applicable

Corrosion (salt spray) Method 1001 of MIL-STD-1344 test condition B

Voltage standing wave ratio (VSWR) From 500 to 18 GHz or approximately 80 percent of upper cutoff frequency of the cable whichever is lower

Cable group II 1 15 + 02 F (GHz)
Cable group VI 1 15 + 01 F (GHz)

Connector durability 500 cycles minimum at 12 cycles per minute maximum The connector shall meet the mating characteristics and force to engage and disengage requirements

Contact resistance In milliohms maximum

	Initial	After environment
Center contact	3 0	4 0
Outer contact	2 0	Not applicable
Braid to body	0 5	Not applicable

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Dielectric withstanding voltage Method 3001 of MIL STD 1344

Cable group II 750 volts rms at 60 Hz at sea level
Cable group VI 1 000 volts rms at 60 Hz at sea level

Vibration high frequency Method 2005 of MIL-STD 1344 test condition III No discontinuity permitted 1/

Shock Method 2004 of MIL-STD-1344 test condition F No discontinuity permitted 1/

Temperature cycling Method 1003 of MIL-STD-1344 test condition A, except test high temperature shall be +200°C for connectors using +200°C cables (see table I) Low test temperature shall be -65°C

Moisture resistance Method 1002 of MIL-STD-1344 test condition B No measurements at high humidity Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity

Corona level (at 70 000 feet) 1/

Cable group II 190 volts rms minimum
Cable group VI 250 volts rms minimum

RF high potential withstanding voltage 1/

Voltage and frequency (at a frequency from 5 to 7 5 MHz)

Leakage current Not applicable

Cable group II 500 volts rms minimum
Cable group VI 670 volts rms minimum

Cable retention force See table II

Coupling mechanism retention force Not applicable

Shielding effectiveness Method 3008 MIL-STD-1344 -85 dB minimum 1/

RF transmission loss (dB maximum) 1/ 03 $\sqrt{\text{frequency GHz}}$ Test at a frequency between 9 and 12 4 GHz

Porosity Applicable

Thermal vacuum outgassing Applicable to space grade connectors only

Flammability Applicable to space grade connectors only

Odor Applicable to space grade connectors only

Toxicity (offgassing) Applicable to space grade connectors only

Safety wire hole pullout Not applicable

Part or Identifying Number (PIN) M31031/01 (see table I)

Group qualification See table III

1/ This test needs to be performed only during initial qualification so long as the qualifying design does not change

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NOTES

1. If a connector manufacturer produces a connector that meets all the requirements for two or more connector PIN s (within the same series and of the same environmental grade) the manufacturer may receive qualification approval for two or more connector PIN s qualifying the one connector It is not necessary that such connectors be in the same group Each connector however must be marked with its own PIN For group qualification the connectors must be of similar design

2. For qualification retention where more than one part is listed in a group in this column data may be supplied on any of those parts in order to retain qualification for those parts in the corresponding right hand column The part does not necessarily have to be the part initially qualified

Supersession This specification supersedes DESC drawing 85073 when a QPL source becomes available

CONCLUDING MATERIAL

Custodians
Army - CR
Navy EC
Air Force 85

Review activities
Army - AT AV EA MI
Navy - AS MC OS SH
Air Force - 17 19 99

Preparing activity
DLA - ES

(Project 5935-3987 01)

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